AMATH/PMATH 331: Applied Real Analysis Fall 2026

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[1] Let $A \subseteq \mathbb{R}^n$ be a non-empty subset, and let $\mathrm{Map}(A, \mathbb{R}^m)$ be the space of maps from A to $\mathbb{R}^m$.

[a] Let (f_k) be a sequence in $\mathrm{Map}(A, \mathbb{R}^m)$ which converges uniformly to a *bounded* function f, so $f \in B(A, \mathbb{R}^m)$. Prove that the f_k are *eventually bounded*. That is, prove that there exists $N \in \mathbb{N}$, such that $k \geq N$ implies $f_k \in B(A, \mathbb{R}^m)$.

Answer:

Proof. Since f_k converges uniformly to f:

$$\forall \varepsilon > 0 : \ \exists N \in \mathbb{N} : \ \forall k \geq N : \ \|f_k(x) - f(x)\| < \varepsilon \quad \forall x \in A$$

So take $\varepsilon = 1$ and the respective N. Also, since f is bounded, take M such that $\forall x \in A : \ \|f(x)\| \leq M$
Now,

$$\begin{aligned} \|f_k(x)\| &= \|f_k(x) - f(x) + f(x)\| && \forall x \in A \\ &\leq \|f_k - f(x)\| + \|f(x)\| && \forall x \in A \\ &< \varepsilon + M = 1 + M && \forall k \geq N, \forall x \in A \end{aligned}$$

Therefore, $\forall k \geq N : \ f_k \in B(A, \mathbb{R}^m)$ □

[b] Suppose (f_k) and (g_k) are two sequences in $\mathrm{Map}(A, \mathbb{R}^m)$ which converge uniformly to *bounded* functions $f, g \in B(A, \mathbb{R}^m)$. Prove that the sequence $(\langle f_k, g_k \rangle)_{k \in \mathbb{N}}$ in $\mathrm{Map}(A, \mathbb{R})$ converges uniformly to $f \cdot g \in \mathrm{Map}(A, \mathbb{R})$, where $\langle \cdot, \cdot \rangle$ is the dot product in $\mathbb{R}^m$.

Answer:

Proof. We have to show

$$\forall \varepsilon > 0 : \ \exists N \in \mathbb{N} : \ \forall k \geq N : \ |\langle f_k, g_k \rangle (x) - \langle f, g \rangle (x)| < \varepsilon \quad \forall x \in A$$

By 1[a], g_k is eventually bounded, so let N_0 such that $\forall k \geq N_0 : \ \|g_k(x)\| \leq M_0$.
Let M_f, M_g be the bounds of f, g.
Let arbitrary $\varepsilon > 0$. Let $\varepsilon' = \frac{\varepsilon}{M_0 + M_f}$. Let N_f, N_g such that:

$$\forall k \geq N_f : \ \|f_k(x) - f(x)\| < \varepsilon' \quad \forall x \in A$$

$$\forall k \geq N_g : \ \|g_k(x) - g(x)\| < \varepsilon' \quad \forall x \in A$$

Take $N = \max(N_f, N_g, N_0)$ and take $k \geq N$.
Now,

$$\begin{aligned} |\langle f_k, g_k \rangle (x) - \langle f, g \rangle (x)| &= |\langle f_k, g_k \rangle (x) - \langle f, g_k \rangle (x) + \langle f, g_k \rangle (x) - \langle f, g \rangle (x)| \\ &= |\langle f_k - f, g_k \rangle (x) + \langle f, g_k - g \rangle (x)| \\ &\leq |\langle f_k - f, g_k \rangle (x)| + |\langle f, g_k - g \rangle (x)| \\ &\leq \|(f_k(x) - f(x))\| \|g_k(x)\| + \|f(x)\| \|(g_k(x) - g(x))\| \\ &< \varepsilon' \|g_k(x)\| + \varepsilon' \|f(x)\| \\ &< \varepsilon'(M_0 + M_f) = \frac{\varepsilon}{M_0 + M_f}(M_0 + M_f) = \varepsilon \end{aligned}$$

□

[c] Show that the assumption of boundedness of the limits in part [b] is essential as follows. Show that the sequence (f_k) in $M(\mathbb{R},\mathbb{R})$ given by $f_k(x) = x + \frac{1}{k}$ converges uniformly on $\mathbb{R}$ to the unbounded function $f(x) = x$, but that $f_k^2 \in M(\mathbb{R},\mathbb{R})$ does not converge uniformly to f^2.

Answer:

Proof. Let arbitrary $\varepsilon > 0$. Take $N > \frac{2}{\varepsilon}$

$$\begin{aligned} |f_k(x) - f(x)| &= \left| x + \frac{1}{k} - x \right| = \left| \frac{1}{k} \right| \\ &\leq \frac{1}{N} < \frac{\varepsilon}{2} < \varepsilon \qquad\qquad \forall k \geq N, \forall x \in \mathbb{R} \end{aligned}$$

So f_k converges uniformly to f.

Now, We have to show

$$\exists \varepsilon > 0 : \ \forall N \in \mathbb{N} : \ \exists k \geq N : \ \exists x \in \mathbb{R} : \ \left| f_k^2(x) - f^2(x) \right| \geq \varepsilon$$

Take $\varepsilon = 1$. Let arbitrary $N \in \mathbb{N}$. Take $k = N$. Take $x = k$

$$\begin{aligned} \left| f_k^2(x) - f^2(x) \right| &= \left| x^2 + \frac{2x}{k} + \frac{1}{k^2} - x^2 \right| = \left| \frac{2x}{k} + \frac{1}{k^2} \right| \\ &= \left| 2 + \frac{1}{N^2} \right| > 2 > 1 = \varepsilon \end{aligned}$$

So f_k^2 does not converge uniformly to f^2. □

[2] Let $f \in \mathrm{Map}(\mathbb{R}^n, \mathbb{R}^m)$ be a map, and suppose that f is *continuous* at $\mathbf{x}_0 \in \mathbb{R}^n$.

Define a sequence (f_k) in $\mathrm{Map}(\mathbb{R}^n, \mathbb{R}^m)$ by $f_k(\mathbf{x}) = f(\mathbf{x}_0 + \frac{1}{k}\mathbf{x})$. Let $g \in \mathrm{Map}(\mathbb{R}^n, \mathbb{R}^m)$ be the *constant map* $g(\mathbf{x}) = f(\mathbf{x}_0)$ for all $\mathbf{x} \in \mathbb{R}^n$.

Prove that, for any $R > 0$, the sequence (f_k) converges uniformly to g on the set $B_R(\mathbf{0})$.

Answer:

Proof. We have to show:

$$\forall R > 0 : \ \forall \varepsilon > 0 : \ \exists N \in \mathbb{N} : \ \forall k \geq N : \ \|f_k(x) - g(x)\| < \varepsilon \quad \forall x \in B_R(0)$$

Let arbitrary $R > 0$ and $\varepsilon > 0$.

$$\|f_k(x) - g(x)\| = \left\| f(x_0 + \frac{1}{k}x) - f(x_0) \right\|$$

Since f is continuous at x_0, there exists δ such that

$$\|y - x_0\| < \delta \implies \|f(y) - f(x_0)\| < \varepsilon \qquad \forall y \in \mathbb{R}^n$$

Now take $y = x_0 + \frac{1}{k}x$.

$$\|y - x_0\| = \left\| \frac{1}{k}x \right\| = \frac{1}{k}\|x\|$$

So now we have

$$\frac{1}{k}\|x\| < \delta \implies \|f_k(x) - g(x)\| < \varepsilon \qquad \forall x \in \mathbb{R}^n$$

Now, if we restrict x to $B_R(0)$, we have $\|x\| < R$, so

$$\frac{1}{k}\|x\| < \frac{R}{k}$$

Therefore, we need to choose N such that

$$\forall k \geq N : \ \frac{R}{k} \leq \delta$$

So take $N = \frac{R}{\delta}$. Then, we have:

$$\forall k \geq N : \ \frac{1}{k}\|x\| < \frac{R}{k} \leq \delta \quad \forall x \in B_R(0)$$

And therefore:

$$\forall k \geq N : \ \|f_k(x) - g(x)\| < \varepsilon \quad \forall x \in B_R(0)$$

So (f_k) converges uniformly to g on $B_R(0)$. □

[3] For $k \in \mathbb{N}$, define $f_k : [-1,1] \to \mathbb{R}$ by $f_k(x) = (1-x^2)^k$.

[a] For any $\delta \in (0,1)$, prove that the sequence of functions (f_k) *converge uniformly* on the subset $[-1,-\delta] \cup [\delta,1]$ of $\mathbb{R}$.

Answer:

Proof. Let $f : [-1,1] \to \mathbb{R}$ be the pointwise limit of f_k. That is:

$$f(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{else} \end{cases}$$

Let arbitrary $\delta \in (0,1)$, $\varepsilon > 0$. Let $A = [-1,-\delta] \cup [\delta,1] \subset \mathbb{R}$
We will show f_k converges uniformly to f on A
We have to show:

$$\exists N \in \mathbb{N} : k \geq N \implies |f_k(x) - f(x)| < \varepsilon \quad \forall x \in A$$

Since $x \in A$, so $x \neq 0$:

$$|f_k(x) - f(x)| = |f_k(x) - 0| = |f_k(x)|$$

Since $x \in A$, we have:

$$\begin{aligned} &0 < \delta \leq |x| \leq 1 \\ &0 < \delta^2 \leq x^2 \leq 1 \\ &0 \leq (1-x^2) \leq 1 - \delta^2 < 1 \\ &0 \leq (1-x^2)^k \leq (1-\delta^2)^k < 1 && \forall k \in \mathbb{N} \\ &|f_k(x)| \leq (1-\delta^2)^k < 1 && \forall k \in \mathbb{N} \end{aligned}$$

So we want

$$(1-\delta^2)^k < \varepsilon$$

Taking logaithms, we have:

$$k \ln\left(1-\delta^2\right) < \ln(\varepsilon)$$

$$k > \frac{\ln(\varepsilon)}{\ln(1-\delta^2)}$$

Therefore, take N large enough such that $N > \frac{\ln(\varepsilon)}{\ln(1-\delta^2)}$.
Now, $k \geq N \implies |f_k(x) - f(x)| < \varepsilon \quad \forall x \in A$. So (f_k) converges uniformly to f on A □

[b] Prove that the sequence of functions (f_k) *does not converge uniformly* on the subset $[-1,1]$ of $\mathbb{R}$.

Answer:

Proof. We need to show that:

$$\exists \varepsilon > 0 : \ \forall N \in \mathbb{N} : \ \exists k \geq N : \ \exists x \in [-1, 1] : \ |f_k(x) - f(x)| \geq \varepsilon$$

Take $k = N$, $\varepsilon = \frac{1}{2}$, $x = \sqrt{1 - (\frac{3}{4})^{\frac{1}{k}}}$ and let arbitrary $N \in \mathbb{N}$.
$x \neq 0$, so $f(x) = 0$.

$$|f_k(x) - f(x)| = \frac{3}{4} > \frac{1}{2} = \varepsilon$$

□

[4] Let $A \subseteq \mathbb{R}^n$ be non-empty. Let (f_k) be a sequence in $\text{Map}(A, \mathbb{R}^m)$ which converges *uniformly* to a limit $f \in \text{Map}(A, \mathbb{R}^m)$. We proved in class that if every f_k is continuous on A, then f is also continuous on A.

[a] Adapt that proof to show that if every f_k is *uniformly* continuous on A, then f is also *uniformly* continuous on A.

Answer:

Proof. We need to show f is uniformly continuous on A.
Let arbitrary $\varepsilon > 0$. We need to show:

$$\exists \delta > 0 : \ \forall x, y \in A : \ \|x - y\| < \delta \implies \|f(x) - f(y)\| < \varepsilon$$

Since (f_k) converges uniformly to f, there exists $N \in \mathbb{N}$ such that:

$$k \geq N \implies \|f_k(p) - f(p)\| < \frac{\varepsilon}{3} \quad \forall p \in A$$

Since f_N is uniformly continuous on A, there exists $\delta > 0$ such that

$$x, y \in A \wedge \|x - y\| < \delta \implies \|f_N(x) - f_N(y)\| < \frac{\varepsilon}{3}$$

Therefore, for all $k \geq N$ and $x, y \in A$ such that $\|x - y\| < \delta$:

$$\begin{aligned} \|f(x) - f(y)\| &= \|(f(x) - f_N(x)) + (f_N(x) - f_N(y)) + (f_N(y) - f(y))\| \\ &\leq \|f(x) - f_N(x)\| + \|f_N(x) - f_N(y)\| + \|f_N(y) - f(y)\| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \qquad < \varepsilon \end{aligned}$$

therefore, f is uniformly continuous on A. □

[b] Explain why we do not need to do any extra work if A is compact. That is, why does part [a] follow immediately if A is compact?

Answer:
Since every f_k is uniformly continuous on A, every f_k is also continuous on A.
Since (f_k) converges uniformly to f, we know that f is continuous on A.
Since A is compact, by the Heine-Cantor theorem, f is uniformly continuous on A.

[5] Let $a < b$. Recall the Weierstrass Approximation Theorem: if $f \in C[a,b]$, then there exists a sequence (p_k) in $\mathcal{P}[a,b]$ that converges uniformly to f on $[a,b]$.

Suppose that we have proved this theorem for the particular case $[a,b] = [0,1]$. Show that the general case of any $[a,b]$ follows from that.

Hint: Consider the map $T : \mathbb{R} \to \mathbb{R}$ given by $T(x) = a + (b-a)x$.

Answer:

Proof. We can see that $T(x)$ is continuous and strictly increasing, since $b > a$. Also, $T(0) = a$, $T(1) = b$. So T maps $[0,1]$ onto $[a,b]$.

Define $g : [0,1] \to \mathbb{R}$ given by $g(x) = (f \circ T)(x) = f(T(x)) = f(a + (b-a)x)$. Now, applying the Weierstrass Approximation Theorem on g, we get a sequence (q_k) in $P[0,1]$ that converges uniformly to g on $[0,1]$.

Now take the inverse of T, $T^{-1} : [a,b] \to [0,1]$:

$$T^{-1}(x) = \frac{x-a}{b-a}$$

And define a new sequence (p_k) on $[a,b]$, such that $p_k(x) = (q_k \circ T^{-1})(x) = q_k(T^{-1}(x))$.

Each p_k is still a polynomial as it is the composition of a polynomial with an affine map.

Since q_k converges uniformly on $[0,1]$ to g, for every $\varepsilon > 0$, exxists $N \in \mathbb{N}$ such that:

$$k \geq N \implies |q_k(x) - g(x)| < \varepsilon \quad \forall x \in [0,1]$$

So we get:

$$g \geq N \implies |p_k(x) - f(x)| = \left|q_k(T^{-1}(x)) - g(T^{-1}(x))\right| < \varepsilon \quad \forall x \in [a,b]$$

So therefore, (p_k) converges uniformly to f on $[a,b]$ □

[6] Let $f \in C[0,1]$. Suppose that $\int_0^1 x^n f(x)\,dx = 0$ for all integers $n \geq 0$.

[a] Find a way to use the Weierstrass Approximation Theorem to show that

$$\int_0^1 |f(x)|^2\,dx = 0.$$

Hint: Consider the integral of $f(f-p)$ for an appropriately chosen $p \in \mathcal{P}[0,1]$.

Answer:

Proof. Applying the Weierstrass Approximation Theorem on f, we get a sequence (p_k) in $\mathcal{P}[0,1]$ that converges uniformly to f on $[0,1]$
Now, consider the integral:

$$\int_0^1 f(x)(f(x)-p_k(x))dx = \int_0^1 f(x)^2 - f(x)p_k(x)dx = \int_0^1 f(x)^2dx - \int_0^1 f(x)p_k(x)dx$$

Now, lets look at $\int_0^1 f(x)p(x)dx$ for any arbitrary polynomial p.
p can be expaneded into $p = \sum_{i=0}^n a_i x^i$. Therefore:

$$\begin{aligned}\int_0^1 f(x)p(x)dx &= \int_0^1 f(x)(\sum_{i=0}^n a_i x^i)dx \\ &= \sum_{i=0}^n a_i \int_0^1 x^i f(x)dx \\ &= \sum_{i=0}^n a_i \cdot 0 \\ &= 0\end{aligned}$$

so

$$\int_0^1 f(x)(f(x)-p_k(x))dx = \int_0^1 f(x)^2dx$$

Now denote $I = \int_0^1 f(x)^2dx$ and $I_k = \int_0^1 f(x)(f(x)-p_k(x))dx$.
We've shown that $I = I_k$ for all k.
Since f is continuous on a compact set, it is bounded by M. Since (p_k) converges uniformly to f, so for every $\varepsilon > 0$, there exists N such that:

$$k \geq N \implies |f(x)-p_k(x)| < \varepsilon \quad \forall x \in [0,1]$$

$$\begin{aligned}|I_k| &= \left|\int_0^1 f(x)(f(x)-p_k(x))dx\right| \\ &\leq \int_0^1 |f(x)||f(x)-p_k(x)|dx \\ &\leq \int_0^1 M\varepsilon dx && \forall k \geq N \\ &\leq M\varepsilon && \forall k \geq N\end{aligned}$$

So I_k converges to 0. But I_k is constant for all k, and therefore it must be equal to 0 for all k.

Therefore, $I = \int_0^1 f(x)^2 dx = 0$. So

$$\int_0^1 |f(x)|^2 = 0$$

□

[b] Explain why this means that $f(x) = 0$ for all $x \in [0, 1]$.

Hint: Suppose that there exists some $a \in [0, 1]$ where $f(a) \neq 0$, and try to derive a contradiction using continuity of f and properties of integration.

Answer:

Proof. Suppose there exists some $a \in [0, 1]$ where $f(a) \neq 0$. Since f is continuous, there exists $\varepsilon > 0$ such that for all $x \in B_\varepsilon(a)$, $f(x) \neq 0$.
So, $\int_{a-\varepsilon}^{a+\varepsilon} f(x)^2 dx > 0$. On the rest of the interval, since $f(x)^2 \geq 0$, we have $\int_0^{a-\varepsilon} f(x)^2 dx \geq 0$ and $\int_{a+\varepsilon}^1 f(x)^2 dx \geq 0$
Therefore, on the whole interval:

$$\int_0^1 f^2(x) dx = \int_0^{a-\varepsilon} f(x)^2 dx + \int_{a-\varepsilon}^{a+\varepsilon} f(x)^2 dx + \int_{a+\varepsilon}^1 f(x)^2 dx > 0$$

Which contradicts our result from above □

[7] Suppose that $f : [0,1] \to \mathbb{R}$ is *continuously differentiable*. This means that:

- f is differentiable at all points in $[0,1]$, where we define differentiability at the end points using one-sided limits, *and*
- the derivative $f' : [0,1] \to \mathbb{R}$ is continuous on $[0,1]$, so $f' \in C[0,1]$.

Show that there exists a sequence of polynomials (p_k) in $\mathcal{P}[0,1]$ such that (p_k) converges uniformly to f and the sequence (p_k') of derivatives converges uniformly to f'.

Hint: Start with f', and consider the fundamental theorem of calculus.

Answer:

Proof. Apply Weierstrass to f' to get a sequence (q_k) in $\mathcal{P}[0,1]$ that converges uniformly to f'. Construct the polynomials p_k given by:

$$p_k(x) = f(0) + \int_0^x q_k(t)dt$$

Integrating a polynomial gives another polynomial, so p_k is a polynomial. Also,

$$p_k'(x) = q_k(x)$$

So $(p_k'(x))$ converges uniformly to f' on $[0,1]$.

By the fundamental theorem of calculus:

$$f(x) = f(0) + \int_0^x f'(t)dt$$

so

$$\begin{aligned}
|f(x) - p_k(x)| &= \left| f(0) + \int_0^x f'(t)dt - f(0) - \int_0^x q_k(t)dt \right| \\
&= \left| \int_0^x f'(t) - q_k(t)dt \right| \\
&\leq \int_0^x \left| f'(t) - q_k(t) \right| dt \\
&\leq x \left\| f'(t) - q_k(t) \right\|_\infty \quad \forall t \in [0,1] \\
&\leq \left\| f'(t) - q_k(t) \right\|_\infty \quad \forall t \in [0,1]
\end{aligned}$$

Since q_k converges uniformly to f', the right hand side above converges to 0, so therefore p_k converges uniformly to f. □

[8] Let $a > 0$ and $c \in \mathbb{R}$. Consider the map $T : \mathbb{R} \to \mathbb{R}$ given by $T(x) = ax^2 + c$.

[a] Show that, depending on a, c, the map T may have two, one, or zero fixed points.

Answer:

[b] Show that if there is only one fixed point, we cannot classify it as attracting or repelling.

Answer:

[c] Show that if there are two fixed points, one is always repelling, and the behaviour of the other depends on whether ac is greater or less than $-\frac{3}{4}$, and we learn nothing if $ac = -\frac{3}{4}$.

Answer:

[9] Let $\lambda > 0$. Consider the map $T : \mathbb{R} \to \mathbb{R}$ given by $T(x) = x + \frac{\lambda}{\pi} \sin(\pi x)$.

[a] Compute all of the fixed points of T. You should find that there are infinitely many, and they are very simple to describe.

Answer:

[b] Show that "half" of the fixed points are always *repelling*.

Answer:

[c] Find a condition on λ which ensures that the other "half" of the fixed points are *attracting*, and find a different condition on λ which ensures that this other half is repelling.

Answer:

[d] For what value of λ do we know nothing about this other half of the fixed points?

Answer:

[10] Let $\lambda > 0$. Consider the map $T : \mathbb{R} \to \mathbb{R}$ given by $T(x) = x + \frac{\lambda}{2\pi} \sin^2(\pi x)$.

[a] Compute all of the fixed points of T. You should find that there are infinitely many, and they are very simple to describe.

Answer:

[b] What can you say qualitatively about the fixed points in this case?

Answer: